

MINOR PROJECT REPORT

Exploratory Data Analysis (EDA)

on the Netflix Movies and TV Shows Dataset

Data Science & Business Analytics Program

Assigned by: Mr. Ayush Shrivastav (Program Mentor)

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Roll Number	DSBA-T12/JY-11157
Project Type	Minor Project - Exploratory Data Analysis
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn (Google Colab)

1. Introduction & Objective

Netflix is one of the world's largest streaming platforms, hosting thousands of Movies and TV Shows across genres, countries, and languages. Understanding how this content library is structured can reveal valuable insights into content strategy, regional focus, and audience targeting. This Minor Project applies the Exploratory Data Analysis (EDA) workflow — data cleaning, preprocessing, and visualization — to the Netflix Movies and TV Shows dataset.

Objective: To perform an in-depth Exploratory Data Analysis on the Netflix dataset in order to clean and preprocess the raw data, engineer useful features, and uncover meaningful patterns, trends, and relationships across content type, genre, country, content rating, release year, and duration through statistical summaries and visualizations.

2. Dataset Description

Dataset Name: Netflix Movies and TV Shows

Source: Kaggle (<https://www.kaggle.com/datasets/shivamb/netflix-shows>)

Total Records: 8,800+ titles | Total Original Columns: 12

Key columns in the dataset:

- show_id – unique identifier for each title
- type – Movie or TV Show
- title, director, cast – descriptive/text fields
- country – country (or countries) of production
- date_added – date the title was added to Netflix
- release_year – original release year of the title
- rating – content/age rating (e.g., TV-MA, PG-13)
- duration – movie length in minutes, or number of TV seasons
- listed_in (renamed to genre) – comma-separated genre categories
- description – short plot summary (dropped during cleaning, as it was not needed for quantitative analysis)

3. Tools & Libraries Used

- Google Colab – cloud-based Jupyter notebook environment used to write and run the project
- Pandas – data loading, cleaning, transformation and aggregation
- NumPy – numerical operations and array handling
- Matplotlib – base plotting engine for charts
- Seaborn – statistical visualizations (count plots, histograms, heatmaps, boxplots) with improved styling

4. Methodology

4.1 Data Import

The dataset was loaded into a Pandas DataFrame using `pd.read_csv()`, followed by an initial inspection using `.head()`, `.tail()`, `.shape`, `.info()` and `.describe()` to understand the structure, data types, and summary statistics of the columns.

4.2 Data Cleaning

1. Checked missing values in every column using `.isnull().sum()` and calculated the percentage of missing data per column.
2. Checked and confirmed there were no duplicate rows using `.duplicated().sum()`.
3. Filled missing 'director' entries with 'Unknown' and missing 'cast' entries with 'Not Available', since these are text/categorical fields where dropping rows would lose valuable data.
4. Filled missing 'country' and 'rating' values using the mode (most frequent value) of each respective column.
5. Filled missing 'date_added' values with 'Unknown' before type conversion.
6. Renamed the column 'listed_in' to 'genre' for clarity.
7. Dropped the 'description' column, as it was not required for the numerical/categorical analysis performed in this project.
8. Exported the intermediate cleaned dataset to 'cleaned_netflix_titles.csv'.

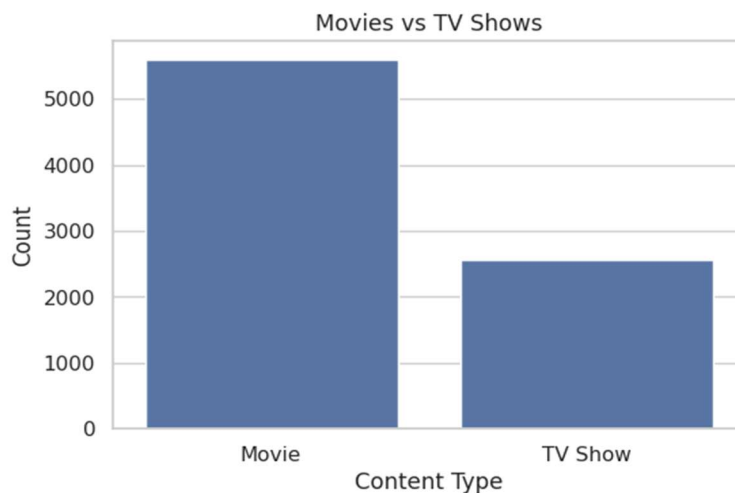
4.3 Data Preprocessing

9. Converted 'date_added' to a proper datetime format using `pd.to_datetime()` with `errors='coerce'` to safely handle any inconsistent date strings.
10. Engineered two new features from `date_added`: 'year_added' and 'month_added', to support year-wise and month-wise trend analysis.
11. Extracted the numeric portion of the 'duration' column (which contained mixed text such as '90 min' or '2 Seasons') into a new numeric column 'duration_num' using regex extraction and `pd.to_numeric()`.
12. Handled any remaining nulls in the engineered columns — 'year_added' filled with 0 and 'month_added' filled with 'Unknown' — then converted 'year_added' to integer type.
13. Re-verified data types and the shape of the final dataset before moving to visualization.

5. Exploratory Data Analysis – Visualizations

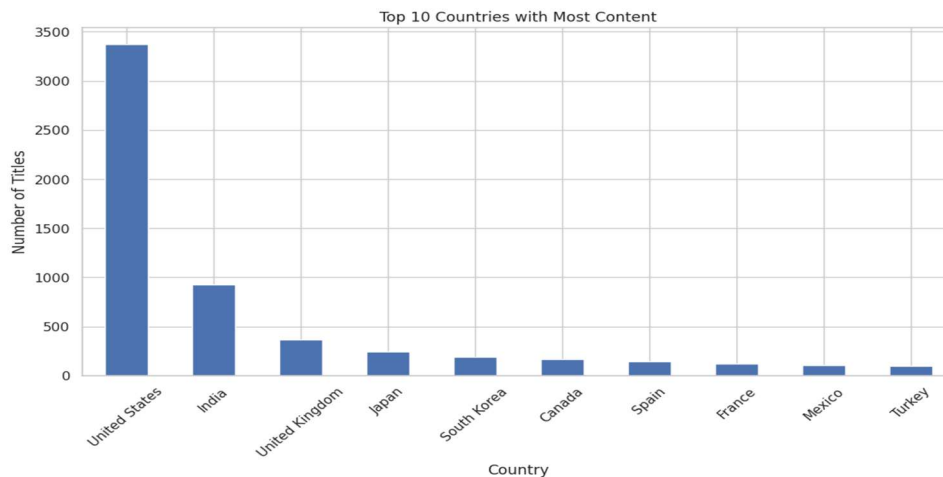
A total of 12 visualizations were created to analyze the dataset from multiple angles — content-type distribution, geographic spread, ratings, release trends, genre popularity, duration patterns, and correlations.

5.1 Movies vs TV Shows (Count Plot)



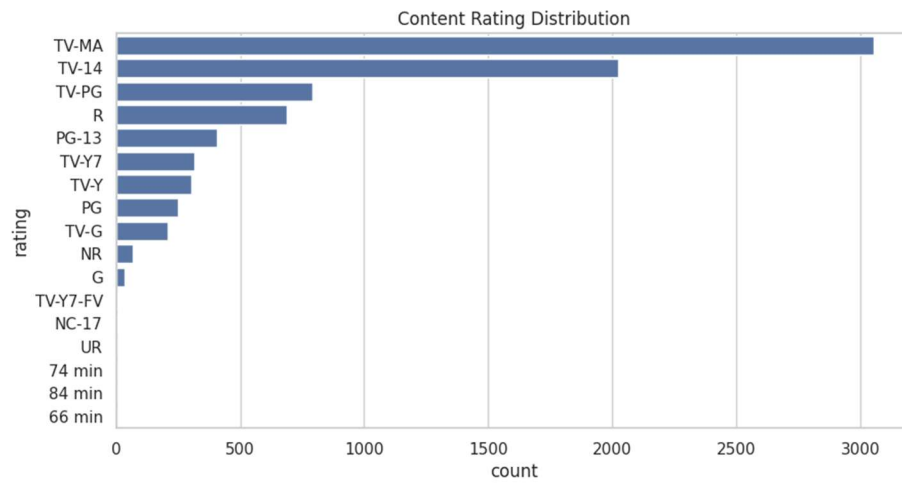
Observation: Movies significantly outnumber TV Shows in the Netflix catalog, indicating the platform's continued heavier investment in film content.

5.2 Top 10 Countries with Most Content (Bar Chart)



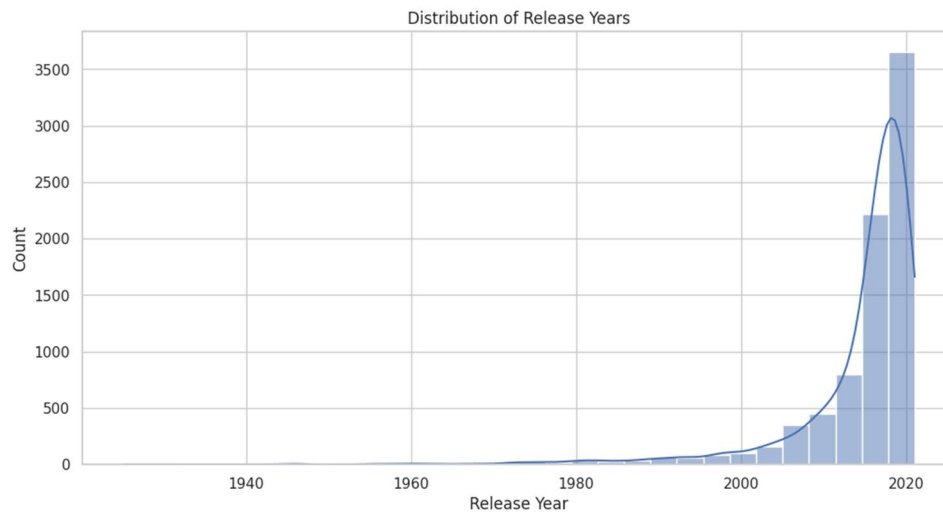
Observation: The United States leads by a wide margin in content production, followed by India and the United Kingdom, reflecting Netflix's strongest content hubs.

5.3 Content Rating Distribution (Horizontal Bar)



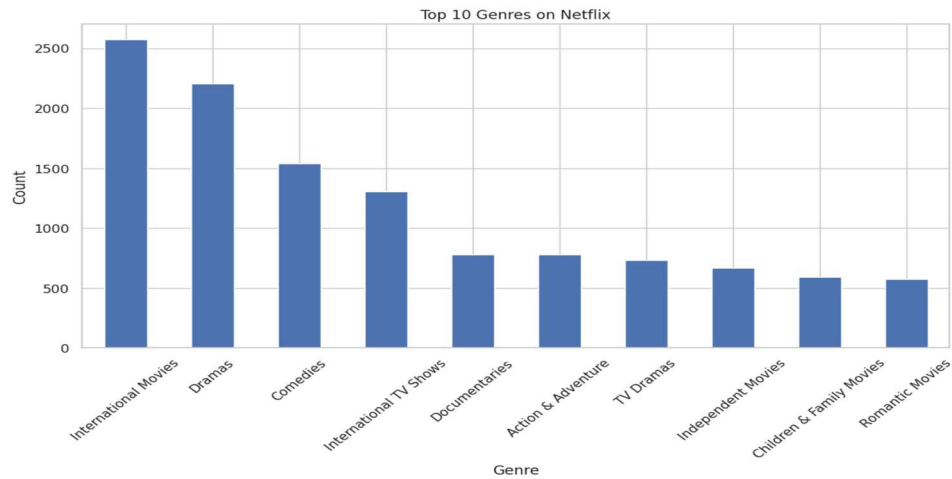
Observation: TV-MA (mature audience) and TV-14 are the most frequent ratings, showing that Netflix's catalog skews toward mature and teen audiences rather than young children.

5.4 Distribution of Release Years (Histogram + KDE)



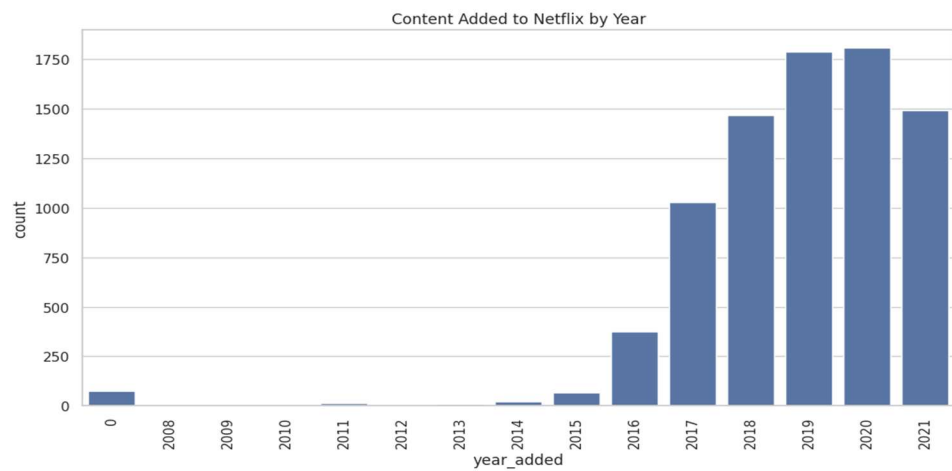
Observation: The distribution is heavily right-skewed — the vast majority of titles were released after 2010, with a steep rise in the last decade.

5.5 Top 10 Genres on Netflix (Bar Chart)



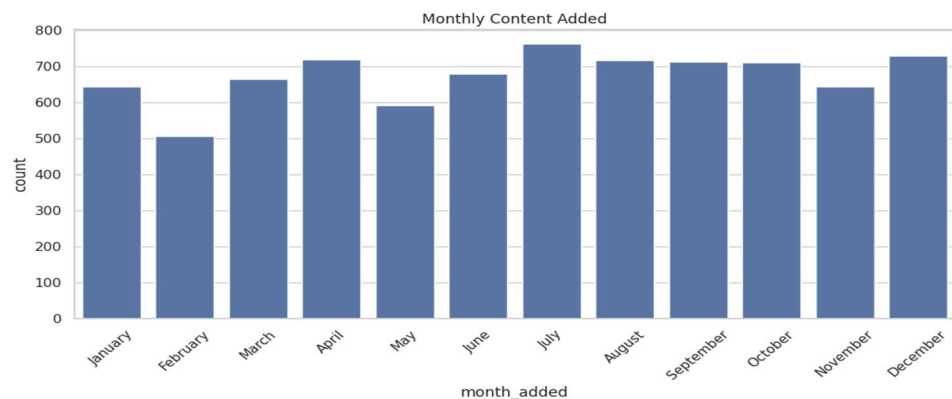
Observation: International Movies, Dramas, and Comedies are the most common genres, showing Netflix's focus on broad, globally appealing categories.

5.6 Content Added to Netflix by Year (Count Plot)



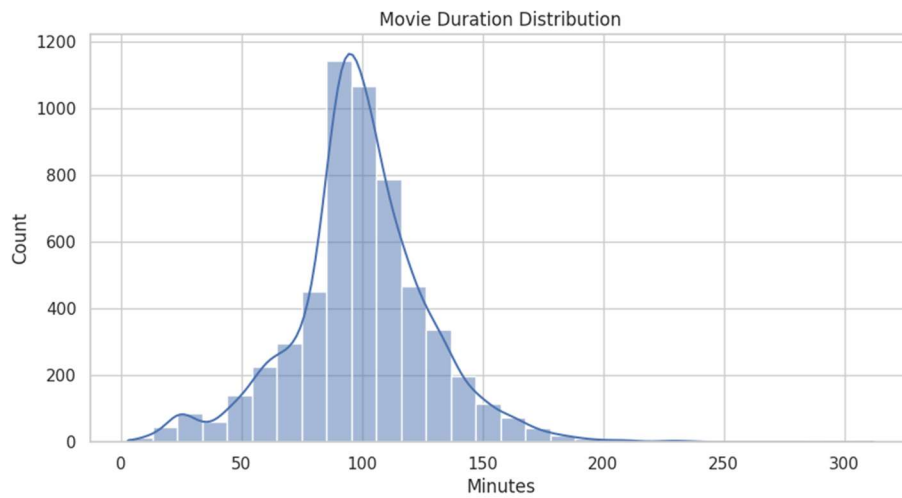
Observation: Content additions grew rapidly between 2016 and 2019, peaking around 2019-2020, before slightly tapering — mirroring Netflix's global expansion phase.

5.7 Monthly Content Added (Count Plot)



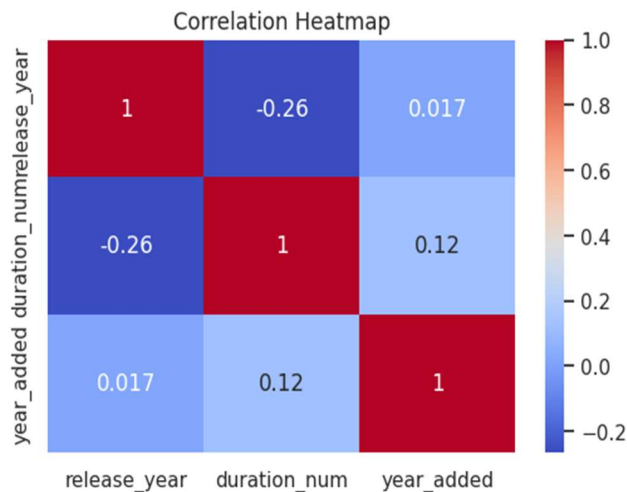
Observation: Content additions are fairly distributed across months, with slightly higher activity toward the end of the year, likely aligned with holiday-season release strategies.

5.8 Movie Duration Distribution (Histogram + KDE)



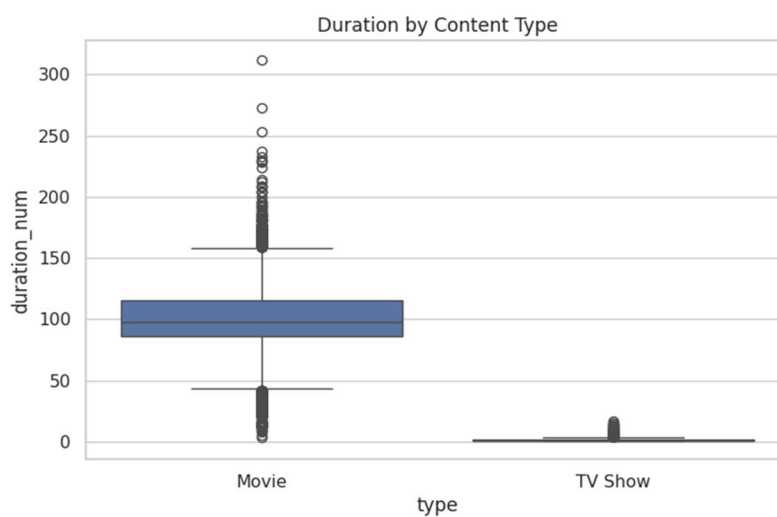
Observation: Most movies fall between 80 and 120 minutes, consistent with standard commercial feature-film runtimes.

5.9 Correlation Heatmap



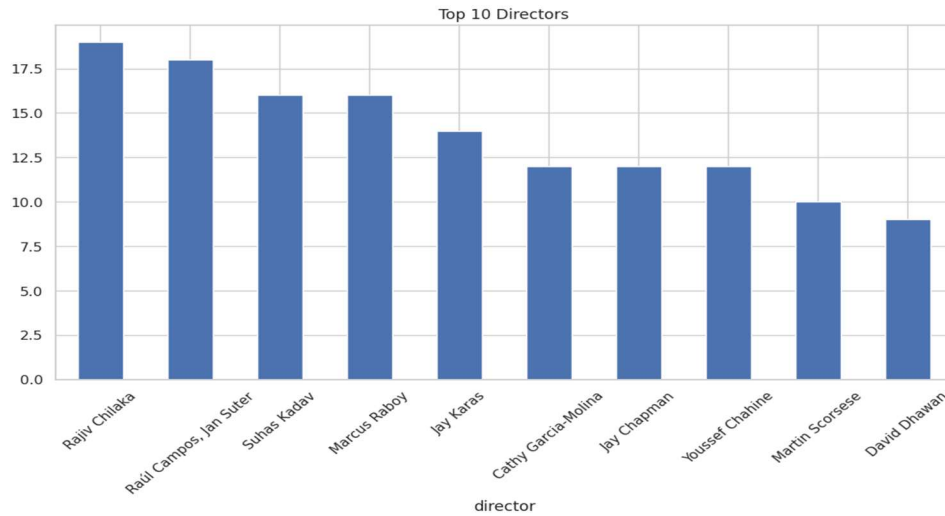
Observation: Release year, duration, and year-added show only weak correlations with each other, suggesting these attributes vary largely independently.

5.10 Duration by Content Type (Boxplot)



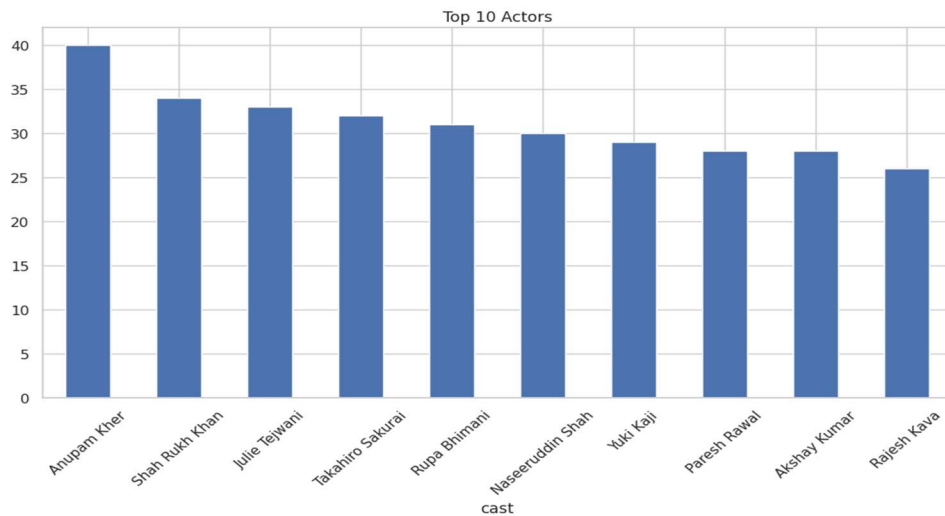
Observation: TV Shows (measured in seasons) show a much tighter, lower-value spread than Movies (measured in minutes), confirming the two content types need separate duration handling.

5.11 Top 10 Directors (Bar Chart)



Observation: A small number of directors (excluding 'Unknown' entries) have multiple titles on the platform, but no single director dominates the overall catalog.

5.12 Top 10 Actors (Bar Chart)



Observation: Certain actors appear across many titles, particularly within regional cinema (e.g., Indian and Anupam Kher-style prolific character actors), reflecting repeated collaborations with Netflix.

6. Summary of Insights

- Movies dominate the Netflix catalog, far outnumbering TV Shows in total count.
- The United States is the largest content contributor, followed by India and the United Kingdom.
- TV-MA and TV-14 are the most common content ratings, indicating a primarily mature/teen-oriented library.
- The majority of available content was released after 2010, with explosive growth in additions between 2018 and 2020.
- International Movies and Dramas are among the most popular genres across the catalog.
- Most movies run between 80 and 120 minutes, matching standard feature-film length.
- A small number of directors and actors appear repeatedly across multiple titles.
- Release year, duration, and year-added show only weak correlation with one another.
- Monthly content additions are fairly steady throughout the year with a mild year-end increase.
- The raw dataset required substantial cleaning — missing values and inconsistent formats — before reliable analysis could be performed.

7. Challenges Faced and Solutions Implemented

Challenge 1: High volume of missing values

Columns such as director, cast, country, rating, and date_added had a significant percentage of missing values. Solution: Used context-appropriate imputation — placeholder text ('Unknown' / 'Not Available') for identity fields, and mode imputation for country and rating, preserving row count instead of dropping data.

Challenge 2: Mixed-format duration column

The 'duration' column stored two different units in the same text field — minutes for Movies and 'Seasons' for TV Shows — making it non-numeric and unusable for statistical analysis. Solution: Used regex extraction (`str.extract('(\d+)')`) combined with `pd.to_numeric()` to create a clean numeric 'duration_num' column.

Challenge 3: Multi-valued categorical columns

The 'genre' (listed_in) and 'cast' columns often contained multiple comma-separated values in a single cell, which would have under-counted individual categories in a direct frequency count. Solution: Applied `str.split(',').explode()` to break each row into individual genre/actor entries before computing `value_counts()`.

Challenge 4: Inconsistent date formats

The 'date_added' column contained inconsistent string date formats and some missing values. Solution: Converted the column using `pd.to_datetime(errors='coerce')`, which safely converts valid dates and flags unparseable ones as NaT for controlled handling, and derived 'year_added' and 'month_added' features from it.

8. Conclusion

This Exploratory Data Analysis project provided valuable insights into the Netflix Movies and TV Shows dataset. The raw data was cleaned by handling missing values, renaming columns, dropping irrelevant fields, and converting data types; it was then preprocessed by engineering new features such as year_added, month_added, and duration_num. A total of twelve visualizations — count plots, histograms, bar charts, a boxplot, and a correlation heatmap — were used to explore trends in content type, country, genre, rating, release year, and duration. Overall, the project demonstrates how a

structured EDA workflow can transform raw, messy real-world data into clear, actionable insights that could support content strategy and business decision-making on a streaming platform such as Netflix.